CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 4: Expansions

Every formula and result of the chapter on one sheet

1	Square of a sum	$(a+b)^2 = a^2 + 2ab + b^2$
2	Square of a difference	$(a-b)^2 = a^2 - 2ab + b^2$
3	Difference of squares	$(a+b)(a-b) = a^2 - b^2$
4	Product	$(x+a)(x+b) = x^2 + (a+b)x + ab$
5	Sum of the two squares	$(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$
6	Difference of the two squares	$(a+b)^2 - (a-b)^2 = 4ab$
7	Reciprocal form	$a^2 + \frac{1}{a^2} = \left(a + \frac{1}{a}\right)^2 - 2 = \left(a - \frac{1}{a}\right)^2 + 2$
8	Trinomial square	$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$
9	Cube of a sum	$(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
10	Cube of a difference	$(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
11	Sum of cubes	$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
12	Difference of cubes	$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
13	Reciprocal cubes	$a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right)$
14	Three cubes	$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
15	Zero sum	If $a+b+c=0$ then $a^3 + b^3 + c^3 = 3abc$